

Shantanu Shastri

shantanushastri70@gmail.com | (412) 608-7945 | linkedin.com/in/shantanushastri

EDUCATION

Carnegie Mellon University	December 2026
Master of Information Systems Management	GPA: 3.85
Manipal Institute of Technology	August 2022
Bachelor of Electronics and Communications Engineering	GPA: 3.99
<i>Relevant Coursework:</i> Data Structures, Algorithms, Distributed Systems, OOP, System Design	

SKILLS

Languages: Java, Python, C++, TypeScript, JavaScript, SQL, Bash/Shell
Backend & Architecture: Spring Boot, RESTful APIs, Microservices, System Design, FastAPI, Node.js, ETL
Databases & Caching: PostgreSQL, MySQL, Redis, Ignite, NoSQL, Neo4j, Milvus, Sybase, MongoDB
Infrastructure & DevOps: Apache Kafka, Zookeeper, Docker, Kubernetes, Azure, AWS, CI/CD, JUnit, Prometheus

EXPERIENCE

BlackRock	January 2022 – August 2026
<i>AI Engineer Internship</i>	June 2026 – August 2026
Massive Data Pipelines: Built a fault-tolerant backend processing large-scale data using FastAPI, Neo4j, Milvus, PostgreSQL, and Kafka for real-time, stream-based knowledge graph ingestion. Vector-Graph Architecture: Architected a hybrid system linking Milvus embeddings to Neo4j nodes via a PostgreSQL bridge, resolving data inconsistencies and enabling structured search under 100ms. RESTful API & GenAI: Engineered a generative AI platform backed by 10+ rate-limited RESTful APIs handling 2K+ peak RPS, advancing Straight-Through Processing toward 100%. Evaluation & Monitoring: Created offline-online evaluation frameworks to track data drift and confidence scores, leveraging policy optimization to handle 3.5M records and \$14T in cash.	
<i>Software Developer 2 - Associate</i>	January 2025 – July 2025
Microservices & Scale: Architected a Java Spring Boot data framework and TTL Azure microservices, automating 1.5M+ client touchpoints and reducing infrastructure storage costs by 40%. Distributed Caching: Engineered a distributed read-through Ignite NoSQL cache with LRU eviction to address high-latency APIs, scaling concurrent requests and cutting response times from 450ms to 150ms.	
<i>Software Developer 1 - Analyst</i>	July 2022 – December 2024
Distributed Systems: Resolved singleton bottlenecks by engineering a sharded server environment with Zookeeper leader election, distributing load and boosting system throughput to 10K+ RPS. SQL Optimization: Achieved a 37% reduction in PostgreSQL query latency on tables exceeding 50M+ rows by refactoring schemas and leveraging composite indexes for faster data retrieval. GenAI Integration: Spearheaded the end-to-end development of a responsive UI, integrating Azure's LLM and machine learning APIs directly into the user workflow. Rigorous Testing & DevOps: Built comprehensive Azure DevOps CI/CD pipelines with integrated JUnit testing suites for robust validation, reducing deployment time-to-market by 40%.	
<i>Software Developer Internship</i>	January 2022 – June 2022
Application Modernization: Re-architected monolithic Pearl/Tomcat applications into a modular Java REST environment and Angular UI, drastically reducing technical debt and deployment time.	

PROJECTS

High-Performance Modular Booking Architecture
Engineered a decoupled microservices architecture using Apache Kafka, seamlessly handling bursts of 100,000+ real-time requests with minimal latency spikes and containerized Docker deployments.
Zulip Open-Source GenAI Extensions
Built a high-throughput message engine using Tornado, Redis, and RabbitMQ to process live data streams into concise LLM-based summaries with zero message loss and optimized API token usage.